Volume 1–20 Canonical Spec v1.0

Canonical Spec
This page is the consolidated specification for *Symbolic Mechanics* Volumes 1–20.

I. System Stratification of Volumes 1–20

🧱 **Layer summary**

- **Layer 0**: Core Engine (Δ→S→L→R→Exit) + regenerative restart (Δ→S→C→S)
- **Layer 1**: Symbol Economy (4-seat distribution)
- **Layer 2**: Axis Geometry (M/F/J, Symbolic Parents, Parental Volume, C-Module emergence)
- **Layer 3**: Boundary Physics (R_s, V, G, A_th, Spotlight / boundary-light, blackout, Orb)
- **Layer 4**: Interaction Field (ΔV, E_h, E_x, attraction-tension, intrusion-resistance dynamics)
- **Layer 5**: Projection Runtime (Prj, TS, M_3, H_p, V_p, B_r, Θ)
- **Layer 6**: Post-Projection Alarm State (alarm takeover after projection shutdown)
- **Layer 7**: Catastrophic Fallback Layer (SB, ShC, Clown, Grand Hall, Firefly)

Layer 0 — Core Engine

Primary cycle

```
Δ → S → L → R → Exit → new Δ
```

Layer 0 contains the canonical processing engine of Symbolic Mechanics. It defines the compulsory movement from perceived difference into symbolic containment, from containment into load, from load into rupture, from rupture into discharge, and from discharge into renewed input. It also includes the processor-selection event that assigns whole-cycle ownership to SCd or SCr.

Regenerative restart (Volume IV)

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Δ → S → C → S
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This loop does not replace the primary engine. It exists inside the engine as a restart path that prevents overload escalation and allows continuation without rupture.

Layer 1 — Symbol Economy

Layer 1 contains the symbolic distribution economy of the system. It defines the four-seat architecture, seat assignment, symbolic weight, replacement resistance, seat-specific holding behaviour, shadow-position pressure, and the persistence logic of installed symbolic objects.

Layer 2 — Axis Geometry

Layer 2 contains the directional geometry of the system. It defines the M-axis, F-axis, and J-axis as load-bearing vectors that bend the route of symbolic processing. It also contains the canonical formalization of Symbolic Parents and parental volume.

Symbolic Parents are not representations of actual parents. They are the system’s two built-in regulatory channels:

- soothing capacity on the M-axis
- boundary-setting capacity on the F-axis

Parental volume determines whether load dissipates, accumulates, redirects, or becomes eligible for C-Module emergence.

When M-axis volume and F-axis volume reach stable mid-range activation, and J-axis dominance naturally falls below override strength, this layer produces the Companion module as a structural stabilizer. The C-Module therefore belongs to core canonical architecture and not to a secondary repair description.

Layer 3 — Boundary Physics

Layer 3 contains the physical rules of access, thresholding, illumination, and shutdown. It defines safety radius, visibility, gate openness, alarm threshold, spotlight or boundary-light, blackout state, Orb formation, attentional capture, boundary override, and shutdown corridor dynamics.

Layer 4 — Interaction Field

Layer 4 contains the dynamic field behaviour that arises when boundary conditions meet external presence. It governs attraction-tension, instruction resistance, intrusion pressure, displacement, seat-field contraction, visibility collapse, sorting failure, existence compensation, visibility differential, exit-homology amplification, and forward-field amplitude modulation.

Layer 5 — Projection Runtime

Layer 5 contains the intimate projection runtime. It defines projector activation, single-input mode, Table-Surface stabilization, Position-3 melt, projector heat, vapor pressure, Bartender reduction, projection threshold, and projection thermal overload.

Layer 6 — Post-Projection Alarm State

Layer 6 contains the state that follows projection shutdown. It formalizes the transfer from projector-owned runtime into alarm-owned runtime.

Layer 7 — Catastrophic Fallback Layer

Layer 7 contains the deepest fallback architecture of Volumes 1–20. It governs shame-break, Shadow-Child formation, Clown generation, Grand Hall scene selection, and Firefly navigation.

II. Complete Variable Table

Variable	Definition
Δ	Input packet of difference. Engine processing begins only after conversion into Δ.
S	Symbolic container generated to hold unresolved difference.
seat	Installed position of S within the 4-seat economy. Each seat imposes distinct load behaviour and routing bias.
L	Accumulated load within S.
R	Rupture threshold. Carrying limit of the installed symbolic container.
Exit	Rupture-triggered discharge pathway. Canonical Exit has 3 forms only: violent, delayed, mourning/absorptive.

Variable	Definition
proc	Cycle owner processor. proc $\in$ {SCd, SCr}. Mutually exclusive during a given cycle.
κ	Processor sensitivity / competitive activation strength. Determines which processor wins access to Δ.
T_rate	Rate of load accumulation, dL/dt.
T_margin	Proximity to rupture, L/R.
W	Symbolic weight.
R_r	Replacement resistance against displacement/substitution.
P = (M, F, J)	Tri-axial geometry: maternal curvature, paternal curvature, Judge curvature.
SP	Symbolic Parents: internal availability of soothing and boundary-setting functions (not biographical parents).
PV	Parental volume: total available regulatory capacity supplied by Symbolic Parents.
M_vol / F_vol	Effective maternal / paternal volume.
C	Companion module (C-Module): self-generated stabilizer enabling regenerative restart.
C_path / LEP / R_cycle	Regenerative restart routing, low-energy path, and $\Delta \to S \to C \to S$ cycle.
O-vector	Organismic movement vector toward displacement/reconfiguration; not intention or narrative wish.
R_s	Safety radius: protected radius inside which approach is processed as critical proximity.
V	Visibility: readable capacity of the symbolic room.
G	Gate openness: access timing/permeability parameter, distinct from V.
A_th / A	Alarm threshold and alarm module.
I°	Intrusion-qualified input (external force recoded into intrusion geometry).
D / C_s / ΔL_i	Displacement vector, seat-field contraction, intrusion-added load.
Spotlight / B_light	Boundary illumination function and boundary-light readability state.
Bk	Blackout state (illumination collapse).
Orb / AC / SCorr	Dark-room anchor, attentional capture, shutdown corridor.

Variable	Definition
B_0 / B_1	Boundary override shutdown / boundary-light reactivation.
ΔV	Visibility differential, $\lvert V\_self - V\_other \rvert$.
E_h / E_x	Exit-structure homology / existence-compensation load.
Prj / Π	Projection state and projection-entry threshold.
TS / M_3 / H_p / V_p / B_r / Θ	Table-Surface, Position-3 melt, projector heat, vapor pressure, Bartender reduction, projection thermal tolerance threshold.
SB / ShC / Clown	Shame-break, Shadow-Child, Clown agency.
scene / Firefly	Active illuminated scene in Grand Hall / lowest-level navigation vector in dark field.

III. Module Table

▼ Core modules (expand)

- **Core Engine**: Δ→S→L→R→Exit→new Δ, plus regenerative restart via C.
- **Processor Module**: SCd vs SCr cycle ownership.
- **Seat Economy**: 4-seat installation, persistence, replacement resistance, shadow pressure.
- **Axis Geometry**: P=(M,F,J), Symbolic Parents, parental volume.
- **Symbolic Parent–Companion Module**: C-Module emergence, Δ→S→C→S restart.
- **Safety-Boundary Module**: R_s, V, G, Spotlight / B_light, A_th.
- **Visibility-Collapse Module**: fog formation, sorting failure, existence compensation.
- **Attraction-Tension Field**: ΔV, R_s, E_h, E_x, O-vector oscillatory dynamics.
- **Intrusion-Alarm Module**: I°→A, D, V↓, C_s, ΔL_i.
- **Shutdown / Blackout Module**: blackout, Orb, AC, SCorr; overload vs voluntary shutdown.
- **Projection Runtime Module**: Prj, TS, M_3, H_p, V_p, B_r, Θ.
- **Post-Projection Alarm Module**: projector collapse → alarm takeover.
- **Shame–Dark Field Module**: SB, ShC, Clown, Grand Hall, Firefly.

IV. Routing Rules

▼ Rules 1–37 (expand)

Basic Routing

1. Every incoming event enters only after conversion into Δ.
2. Processor selection: $proc\_t = argmax \{ κ\_SCd(Δ\_t), κ\_SCr(Δ\_t) \}$.
3. Generate containment S_t for Δ_t.
4. Seat assignment: seat_t = seat_map(proc_t, W_t, context_t).
5. Load: L_(t+1) = L_t + g(seat_t, W_t, P_t, context_t).
6. Rupture: if $L\_t \geq R\_t$, rupture triggers.
7. Exit forms: violent, delayed, mourning/absorptive.

Symbolic Parent–Companion Routing

1. Symbolic Parents routed functionally (M_vol, F_vol).
2. PV increases through functional reliability.
3. C emerges when M_vol and F_vol exceed thresholds and J falls below dominance.
4. C creates restart node routing Δ through C before overload.
5. LEP occurs when $Δ \rightarrow C$ bypasses overload and re-enters S.
6. With C_path, load redirects rather than releases.
7. If C conditions fail, system reverts to rupture-governed routing.

Boundary Routing

1. Open_t = 1 iff $V\_t \geq V\_threshold$ and $G\_t \geq G\_threshold$.
2. Directional intake assigned by dominant differential pull.
3. Intrusion: $I° \rightarrow$A-trigger iff authority(source)=1 and distance $\leq R_s$.
4. Alarm chain: $A \rightarrow D \rightarrow V\downarrow \rightarrow C_s$.
5. Repeated intrusion adds $ΔL_i$.

Shutdown / Blackout Routing

1. Overload blackout when Seat 1 + Seat 3 + Seat 4 forces illumination failure.
2. Blackout: Spotlight=0, B_light=0, V=0, Bk=1.
3. Orb forms automatically at blackout.
4. AC = 1 under Orb dominance.
5. Voluntary shutdown: Seat 2 + Seat 3 + Judge compression → B_0=1, Spotlight=0, SCorr active.
6. Corridor behaviour governed by anchoring, not symbolic evaluation.
7. Corridor ends → B_1=1.
8. After reactivation, delayed load re-enters awareness; shame as delayed mismatch detection.

Projection Routing

1. Prj=1 if ($L_1+L_2+L_4 \geq \Pi$) and Δ present and G softened.
2. Single-input mode; TS online; M_3 only correction channel.
3. $P\_total = (H\_p \times V\_p) + M_3 - B\_r$.
4. Termination: if $P\_total > \Theta$ → shutdown.
5. Projection termination transfers ownership into post-projection alarm state.

Shame–Dark Field Routing

1. Shame-break requires SCr investment + interference authority.
2. Shadow-Child forms after SB.
3. Clown generated when buried shame persists + visibility demand remains unsatisfied.
4. Scene activation follows dark-field navigation.
5. Firefly directs; Clown executes; SCr narrativizes after the shift.

V. Compressed State Transition Map

🧭 **One-page map**

- **Ordinary**: Δ → S → proc/seat → L → R → Exit → new Δ
- **Regenerative**: Δ → S → C → S
- **Boundary opening**: V & G ≥ threshold → Open → Δ directs → V holds
- **Blackout**: Seat1+Seat3+Seat4 collapse → Spotlight=0 → Orb → AC
- **Voluntary shutdown**: Seat2+Seat3+Judge compression → B_0 → Orb corridor → B_1
- **Projection**: Π crossed → Prj → TS + M_3 → H_p/V_p → Θ → shutdown
- **Post-projection**: shutdown → alarm takeover
- **Fallback**: SB → ShC → Clown/Grand Hall → Firefly navigation

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VI. Complete Engine Master Expression

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Engine₍20₎ = Core Engine × Symbol Economy × Axis Geometry × Symbolic Parent–Companion Module × Boundary Physics × Interaction Field × Projection Runtime × Post-Projection Alarm State × Catastrophic Fallback Layer
```

VII. Canonical Consistency Corrections

- **Exit taxonomy**: rupture-triggered Exit has 3 forms only (violent, delayed, mourning/absorptive).
- **Low-energy path**: Δ → S → C → S is a non-rupture restart, not an Exit.
- **Symbolic Parents**: functional regulatory channels, not biographical parents.
- **C-Module**: canonical engine module with strict emergence conditions.
- **Blackout vs voluntary shutdown**: distinct triggers and purposes.
- **Orb naming**: Orb is the canonical name (attractor language absorbed).

- **Projection vs alarm**: distinct regimes; alarm takeover begins immediately after projection termination.
- **Fallback**: SB/ShC/Clown/Grand Hall/Firefly belong to catastrophic continuation architecture.

VIII. Final Conclusion

Across Volumes 1–20, Symbolic Mechanics resolves into a deterministic multi-layer engine in which:

- all incoming difference is converted into Δ
- containment, load, rupture, Exit, and renewal follow canonical routing
- regenerative restart exists under C-Module conditions
- boundary physics and interaction fields govern external engagement
- dark-room states are governed by Orb under illumination failure or override
- projection runtime ends by thermal overload
- post-projection continues under alarm
- catastrophic fallback proceeds through SB → ShC → Clown/Grand Hall → Firefly navigation